

Methods of proof – 6

Contradiction and proofs of impossibility

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Theoretical minimum. Proof by contradiction is a *form*, not a method: it tells you how to begin the sentence, not what to do next. This sheet separates the form from the content and collects the genuine tools for proving that something cannot exist.

Fact 1 (three different things). To prove $A \Rightarrow B$ one may: (i) argue *directly*, from A to B ; (ii) prove the *contrapositive* $\neg B \Rightarrow \neg A$, which is a direct proof of an equivalent statement; (iii) argue *by contradiction*, assuming A and $\neg B$ and deriving a false statement. The second and the third are constantly confused. A useful test: if the contradiction you reach is exactly $\neg A$, you have written a contrapositive proof with an extra negation at the front, and it should be rewritten — on an olympiad an unclear write-up costs points.

Fact 2 (negation with quantifiers). $\neg \forall x \exists y P(x, y)$ is $\exists x \forall y \neg P(x, y)$. Almost every failed proof by contradiction begins with a wrongly formed negation; write it out explicitly before doing anything.

Fact 3 (when contradiction is unavoidable). For statements of the form “no object with property P exists” there is nothing to construct, and the contradiction is honest. For existence statements it is a choice, and a choice with a price: a proof by contradiction produces no example (Methods of proof – 5). The same distinction is the intuitionistic one — excluded middle is what allows $\neg \neg A \Rightarrow A$, and it is exactly the step that destroys the example.

Theorem 1 (the toolkit of impossibility). There are only a few ways to prove that an object cannot exist. *An invariant*: a quantity preserved by every admissible step separates the reachable from the unreachable (Methods of proof – 4). *Parity or a colouring*: the simplest invariant. *Counting or*

dimension: two sides of an equality have different sizes. *Descent*: a hypothetical object produces a strictly smaller one (Methods of proof – 2). *The extremal choice*: take the smallest counterexample and improve it (Methods of proof – 3). *A homomorphism*: map the problem into a smaller structure — reduce modulo n , take a determinant, a degree, a sign, a valuation — where the statement is visibly false.

Theorem 2 (irrationality: the standard scheme). To prove that α is irrational, assume $\alpha = p/q$ and produce an integer N with $0 < |N| < 1$. For $\sqrt{2}$ the descent version is: a smallest representation p/q yields a smaller one. For e the number $N = q!(e - \sum_{k \leq q} 1/k!)$ works. The same scheme with a degree argument shows that $\sqrt[3]{2}$ has degree 3 over $\mathbb{Q}$, whence the classical impossibilities of doubling the cube and trisecting the angle by ruler and compass: a constructible number has degree a power of two.

1. Write out the precise negations of: “every continuous function on $[0, 1]$ attains its maximum”, “for every $\varepsilon > 0$ there is $\delta > 0$ such that ...”; “the sequence (a_n) converges”. Then take the standard proof that $\sqrt{2}$ is irrational and rewrite it twice — once as a contrapositive and once as a descent — and say which of the two contains the mathematics.
2. Prove that there does not exist a ring with exactly five regular elements. (An element a is regular if $ax = 0$ or $xa = 0$ forces $x = 0$; the ring need not be commutative, finite, or unital.) (IMS 2008)
3. Prove that there is no rational function R with real coefficients such that $R(n) = 1 + \frac{1}{2^2} + \cdots + \frac{1}{n^2}$ for infinitely many natural numbers n . (Proposed problems)
4. Prove that a closed disc of radius 1 in $\mathbb{R}^2$ cannot be covered by two closed discs each of radius less than 1. (Высшая лига 2023)
5. Prove that $\log_2 3$ and $\sqrt{2} + \sqrt{3}$ are irrational, and that e is irrational by the $0 < |N| < 1$ scheme. Then state exactly which step of each proof

uses that the assumed object is a *ratio of integers*, and which uses only that it is algebraic.